

Robust calculation of the midsagittal plane in CT scans using the Kullback–Leibler’s measure

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Abstract

Objective The identification of the interhemispheric fissure (IF) is important in clinical applications for brain landmark identification, registration, symmetry assessment, and pathology detection. The IF is usually approximated by the midsagittal plane (MSP) separating the brain into two hemispheres. We present a fast accurate, automatic, and robust algorithm for finding the MSP for CT scans acquired in emergency room (ER) with a large slice thickness, high partial volume effect, and substantial head tilt.

Materials and methods An earlier algorithm for MSP identification from MRI using the Kullback–Leibler’s measure was extended for CT by estimating patient’s head orientation using model fitting, image processing, and atlas-based techniques. The new algorithm was validated on 208 clinical scans acquired mainly in the ER with slice thickness ranging from 1.5 to 6 mm and severe head tilt.

Results The algorithm worked robustly for all 208 cases. An angular discrepancy ($^{\circ}$) and maximum distance (mm) between the calculated MSP and ground truth have the mean value (SD) 0.0258° (0.9541°) and 0.1472 (0.7373) mm, respectively. In average, the algorithm takes 10 s to process of a typical CT case.

Conclusion The proposed algorithm is robust to head rotation, and correctly identifies the MSP for a standard clinical CT scan with a large slice thickness. It has been applied in our several CT stroke CAD systems.

Keywords Midsagittal plane · CT · Symmetry · Stroke

Introduction

The interhemispheric fissure (IF), or also known as the longitudinal cerebral fissure, is one of the significant anatomical landmarks in the human brain. Its identification is important for brain segmentation, registration, symmetry assessment and pathology detection. The IF is usually approximated by the midsagittal plane (MSP), which separates the brain into two hemispheres. With the knowledge of the MSP, one will be able to identify several key landmarks located on the MSP, such as the anterior commissure (AC) and posterior commissure (PC) [1,2] which are important landmarks in brain atlases such as the Talairach–Tournoux atlas [3] and Schaltenbrand–Wahren atlas [4]. After identifying these key landmarks, we can perform atlas to patient data mapping for further analysis of the brain structures in the patient’s volume [5]. In addition, the MSP facilitates comparison of the left and the right hemispheres, allowing the analysis of brain asymmetry for pathology [6–8]. For instance, some studies have found temporal lobe asymmetry as a sign of schizophrenia [9,10]. Another study also shows that neuroanatomic abnormality in schizophrenia is found in several structures on the MSP, such as corpus callosum, brainstem, and cerebellum [11]. In acute ischemic stroke, the MSP facilitates automatic and rapid identification of infarct hemisphere [12]. In CT scans, particularly, the MSP is also used to estimate the midline shift, which is a commonly used quantitative feature

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for evaluating severity of brain compression that can lead to death [13].

Various approaches to compute the MSP have been reported. Several studies compute the MSP by applying principal axes method followed by optimization [14, 15]. Some others construct the MSP by applying intensity-based cross-correlation method to the 2D images to obtain a plane which exhibits the maximum bilateral symmetry [16, 17]. The method [7] identifies the MSP by maximizing bilateral symmetry in the images by applying an edge-based cross-correlation approach for the entire head. It is claimed to be robust to rotation (see Comparison with cross-correlation technique for MSP extraction in “Discussion”). The procedure reported in [18] is based on computation of local similarity measure of two sides of the head and is performed using block matching procedure. In other study, Guillemaud et al. [19] calculate the MSP by identifying the fissure line for each slice using linear snakes and orthogonal regression to construct the MSP. This method faces problem when the fissure line deviates significantly from the straight line due to mass effect.

In our previous works, we have proposed two approaches to calculate the MSP based on local symmetry and Kullback–Leibler’s (KL) measure. The first method [20] uses local symmetry measure to compute the MSP in MR images. This method uses two-stage non-iterative searching strategy to localize the IF region and to identify IF line segments with the maximum local symmetry which are subsequently used to construct the MSP. The second method presented in [21] defines the MSP as a plane with the maximum KL-measure with respect to the reference planes. This approach has been validated for MR images (T1, T2, MRA, SPGR, FLAIR) and tried initially for a few CT scans. This method is further extended for diffusion and perfusion weighted images [22], showing that it may work across different pulse sequences. These two approaches have been incorporated into our MR stroke CAD system [23] trial licensed to several hospitals and companies worldwide.

A common limitation possessed by the previous methods is that most of them have a preliminary assumption that the MSP should be oriented close to a vertical plane [18, 20, 21, 24, 25], except [7]. Therefore, the algorithms break down when the head is tilted more than several degrees, for instance, 30° in method [20] or 7° in coronal direction in method [21]. Additionally, the previously mentioned approaches were mainly validated on MR images [14, 15, 25]. Only a few methods were assessed on CT images, and among these methods, no extensive validation has been performed on CT.

CT scans are usually characterized with big slice thickness as well as large partial volume effect. Additionally in emergency cases, the acquisition is often performed when the patient’s head is arbitrarily positioned. In some cases the

head takes atypical position due to, e.g., simultaneous drug administration. Therefore, the acquired scans may be with a highly tilted head simultaneously in all three directions: yaw (rotation in axial plane), roll (rotation in coronal plane), and pitch (rotation in sagittal plane). Due to the high slice thickness and partial volume effect, the brain structures may not be visible continuously through the successive slices. Even a small head tilt may cause a brain structure to be visible on one side of the brain and unseen on the other side, or to have different manifestation in hemispheres. These typical characteristics of CT pose a challenge to formulate a robust algorithm for MSP calculation in CT scans.

There are several approaches focusing on the MSP calculation specifically on CT scans. They are based on knowledge that the falx cerebri can be used as reference to estimate the MSP. The method [13] extracts the falx cerebri from superior axial slices, estimates the midsagittal line for each slice and constructs the MSP afterwards. Similarly, the method by Darius and Mecislovas [24] calculates midsagittal line from the falx attachment to the skull which is usually characterized by skull thickening in the attachment region. These approaches may suffer from false identification of the falx cerebri or skull thickening. In addition, the falx cerebri may not be visible in the scan, particularly, when the brain is swollen.

Our problem is to devise a method, applicable in a clinical setting, which is able to identify the MSP very accurately, robustly, and rapidly in CT scans acquired in the emergency room (ER). In our opinion, none of the existing methods meets these criteria, including our previous three methods. To solve this problem, we have extended our KL-based MSP method [21] to increase its robustness and accuracy in computing the MSP for CT scans acquired in ER. To do this, the algorithm [21] is extended by including additional steps to calculate the coarse estimation of patient’s head orientation and to adjust the orientation if necessary. These steps include brain extraction (skull and gantry removal), skull-based elliptical fitting, volume computation of left–right hemispheres and volume rotation. To perform these steps, we utilize properties of CT head acquisition, such as the knowledge about the range of intensities for bone and brain tissue available from the DICOM header as well as some atlas knowledge from the Talairach–Tournoux atlas [3]. In addition, we also validate the proposed method on a large set of clinical CT cases and perform a statistical analysis.

Materials and methods

Materials

There are altogether 208 clinical datasets used in this study. The datasets include normal and pathological cases, such as stroke. The slice thickness in the patient’s data varies from

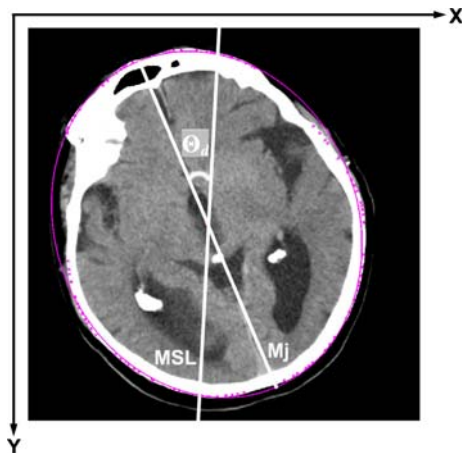


Fig. 1 Axial image with the ellipse fit showing MSL, M_j , and θ_d

1.5 to 6 mm (where 95% of them have slice thickness of 5 mm or more) and the pixel size in the axial plane varies between 0.3906 and 0.5547 mm. For all datasets, the whole head was CT scanned. Brain volume is assumed to be in the radiological convention: the X coordinate axis corresponds to the left–right direction, Y to anterior–posterior direction, and Z to inferior–superior direction.

Method

Algorithm

The KL-based MSP algorithm [21] assumes that the orientation of the MSP is close to a vertical plane or with a minor degree of head tilt. Therefore, in the cases where the head is highly tilted, this algorithm is unable to identify the MSP correctly. In such cases, it is necessary to rotate the scan such that the orientation of the projected MSP on the axial plane is approximately vertical. The angle of rotation is estimated based on fitting an ellipse to the skull outline in the axial plane and estimating head asymmetry.

We studied the performance of method [21] to investigate when the rotation is required. We analyzed the whole datasets and quantified the deviation of the MSP calculated using [21]. Two parameters (θ_d and lr_ratio) were defined to measure the deviation of the MSP. Let MSL be the line obtained from the projection of the MSP to the axial plane and M_j be the major axis of ellipse fitted to the outer edge of the skull in the axial slice with the largest anterior–posterior extent (Fig. 1). Then the first parameter θ_d is defined as the angle between the MSL and M_j . The second parameter enables hemispheric volume comparison and is defined as follows. In the axial plane, let us divide the brain tissue region according to the MSL into two regions. Brain₁ is the region to the left of MSL and brain₂ is the region to the right of MSL. By computing the total area of brain₁ and brain₂ in

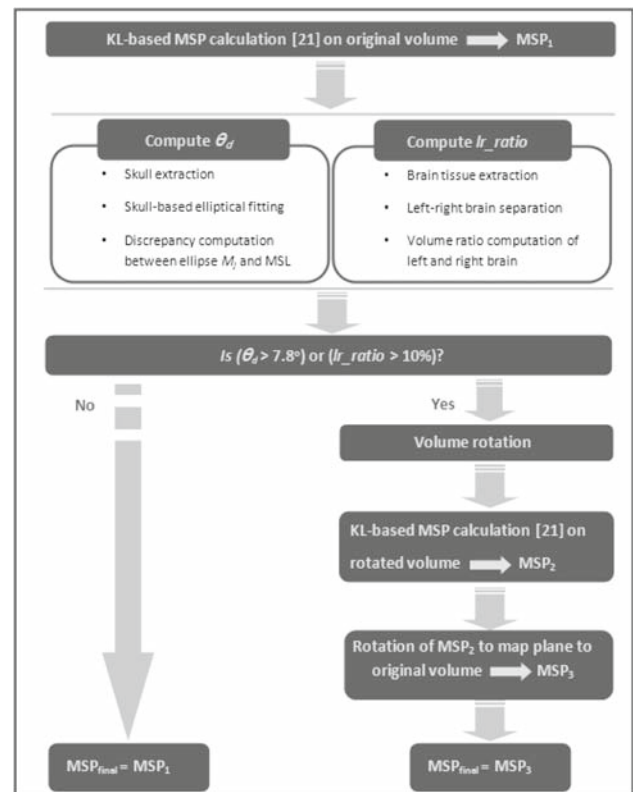


Fig. 2 Flowchart of MSP calculation from CT scans using the KL-measure

each axial slice, the lr_ratio is defined as the ratio between the summation of area of brain₁ across all axial slices and summation of area of brain₂.

Figure 2 presents the flow chart of the proposed method. Initially, the MSP is calculated using KL-measure [21]. Let us denote this as MSP₁. Next, two parameters (θ_d and lr_ratio) are computed to measure the deviation of MSP₁. If (i) θ_d is less than 7.8°, and (ii) lr_ratio is less than 10%, MSP₁ can be considered as accurate and this MSP is taken as the final MSP. Conversely, if (i) θ_d is more than 7.8°, or (ii) lr_ratio is more than 10% (see Parameter justification in “Discussion”), the following steps are performed. The volume is rotated to obtain a new volume with upright IF in the axial view. Algorithm [21] is then re-applied to the new volume to get the second MSP (MSP₂). Finally, based on the previously used rotation angle, the final MSP is obtained by rotating the MSP₂ back so that it coincides with the IF in the original volume.

MSP calculation with KL-measure

Let $p = \{p_i\}$ and $q = \{q_i\}$ be two discrete distributions where p_i and q_i are the probabilities of encountering of the i th state in a set of system states; then the KL-measure is defined as

$$\begin{aligned}
 I(p/q) &= \sum_i p_i \log(p_i) - \sum_i p_i \log(q_i) \\
 &= \sum_i p_i \log(p_i/q_i)
 \end{aligned} \quad (1)$$

Given the brain volume, the MSP is computed in two stages. In the first stage, compute the KL-measure for each slice in a sagittal slab ranging between ± 20 mm from the centroid of the volume, and the slice giving maximum KL-measure is marked as the coarse MSP. In the second stage, a finer search is carried out by perturbing the corner points of the coarse MSP to obtain a perturbed plane with the maximum KL-measure, which is taken as the final MSP.

Calculation of θ_d

Let θ_{MSP} be the angle between MSL and Y axis and θ_{ellipse} angle between M_j and Y axis (positive angle is measured clockwise), Fig. 1. Then θ_d can be calculated as the absolute difference between θ_{MSP} and θ_{ellipse} . The calculation of θ_{MSP} is straightforward and can be computed from the MSL in any axial slice in the volume. The calculation of θ_{ellipse} requires the following few steps:

1. Obtain the reference axial slice (*slice_ref*) to fit the ellipse. This *slice_ref* is the axial slice with the largest anterior–posterior extent which is used to approximate the axial plane passing through the anterior commissure (AC). To find *slice_ref* the following are performed:

- Obtain an initial volume of interest (VOI), which is defined as starting from slice s (approximating location of the AC plane) up to the superior-most slice, where s is calculated as follows:

$$s = \text{round}((43/(43 + 74)) \times \text{total number of axial slices})$$

Values 43 and 74 are derived from the Talairach–Tournoux atlas [3], where 43 mm is the distance from the AC to the most inferior point of the temporal cortex (all acquisitions approximately start from this level) and 74 mm is the distance from the AC to the superior landmark of brain. This is the approximate position for the AC axial plane. According to the Talairach–Tournoux atlas [3], this plane has the maximum anterior–posterior extent. Hence, we can assume that the skull should have the maximum eccentricity and the major axis should approximately correspond to the direction of the IF.

- For each axial slice in the VOI, apply skull-based thresholding with lower and upper thresholds of 350 and 2,250 Hounsfield Unit (HU), respectively, to get a binarized skull. These threshold values were selected based on the information about typical

intensity of bone in CT, which is approximately 1,000–2,250 HU [26] as well as taking into account the partial volume effect.

- Extract the largest connected component for each axial slice in the VOI, and find the anterior–posterior extent of the connected component.
 - The axial slice at which the anterior–posterior extent reaches its maximum is marked as the *slice_ref*.
2. Apply skull-based thresholding on the *slice_ref* and find the largest connected component of the skull.
 3. Obtain the outer edge of the skull component calculated in step 2 and fit the ellipse [27] to the outline of the skull.
 4. Calculate θ_{ellipse} from the ellipse parameters obtained in step 3.

Calculation of lr_ratio

1. Generate a binarized volume by applying thresholding (based on the window center and window width parameters from the DICOM header) to the patient's volume to extract brain tissue and to remove the skull and gantry (typically the values of lower and upper threshold are around -5 and 75 HU, respectively). In the binarized volume, a white voxel represents brain tissue and black one non-brain tissue.
2. Next, the location of each brain tissue voxel with respect to the MSP is analyzed. Let the MSP_1 be defined using a plane equation $Ax + By + Cz + D = 0$ where $A > 0$, and x, y, z are coordinates in patient's volume. For each white voxel with coordinates (v_x, v_y, v_z) in the binarized volume, the value of $f(v_x, v_y, v_z) = Av_x + Bv_y + Cv_z + D$ is calculated. The value of $f(v_x, v_y, v_z)$ can be negative or positive, which implies that the corresponding voxel is located to the right or to the left of MSP_1 , respectively. Thus, the volume of left brain (V_{left}) can be calculated as the total number of white voxels in which $f(v_x, v_y, v_z)$ yielding positive value. Similarly, the volume of right brain (V_{right}) is given by total number of white voxels where $f(v_x, v_y, v_z)$ yielding negative value. Then, the parameter lr_ratio is computed as

$$\frac{|V_{\text{right}} - V_{\text{left}}|}{\max(V_{\text{right}}, V_{\text{left}})}.$$

Volume rotation

The volume rotation is performed as follows. First, the center of mass of the binarized *slice_ref* with $(X_c, Y_c, Z_{\text{slice_ref}})$ coordinates is calculated. This center of mass is projected to each axial slice and then used as the center of rotation for each slice. For example, slice number 5 will be rotated with respect to $(X_c, Y_c, 5)$. Next, each axial slice is rotated by an angle— θ_{ellipse} , where the rotated slice is constructed using bilinear interpolation. This rotation procedure is comparable

to rotating the whole volume with respect to an axis passing through the center of mass of *slice_ref* and orthogonal to the axial plane.

Analysis

To determine the parameters used in the proposed method, eightfold cross validation was performed by dividing the datasets randomly into eight equipotent sets. The leave-one-out method was used such that in each experiment seven groups were used as training set and one group as test set.

For quantitative analysis, the ground truth was marked by neuroanatomy experts (WLN, WA) manually. The ground truth is generated using a dedicated tool which allows the expert to draw the ground truth line (GTL) on the axial slices. The GTL was drawn slice by slice such that they were passing centrally through the IF.

Three measures are computed to perform the quantitative analysis. First, the angular discrepancy (denoted as α , in degrees), which is the average absolute angle between the MSP and each individual GTL. Second, the distance discrepancy (denoted as d , in mm), which is the average Euclidian distance between the end points of GTL and MSP, where the end points of GTL are marked on the bounding box of the brain (to capture the highest error occurred). Third, the angular deviation from planarity (denoted as θ_{GT} , in degrees), which is the maximum angular deviation amongst the GTL and it characterizes the curvature of the IF.

The proposed method is validated against a large set of clinical CT scans acquired in ER. We analyzed the mean and SD of α and d as well as its error distribution. We also performed two correlation tests for checking the existence of correlation between θ_{GT} and α as well as between θ_{GT} and d .

Additionally, the robustness of the method towards rotation was checked using a high-resolution CT scan with 1.5 mm slice thickness with rotation in yaw, pitch, and roll directions. Trilinear interpolation was used to generate rotated volumes. The data were rotated in clockwise direction from 5° to 40° in the yaw direction, from 5° to 30° in the

Table 1 Statistical parameters of distance and angular discrepancies between the calculated MSP and ground truth MSP for clinical CT scans

	α (°)	d (mm)	θ_{GT} (°)
Mean	0.0258	0.1472	0.8751
SD	0.9541	0.7373	0.9330
Max	3.4496	2.5753	5.2995

pitch direction, and from 5° to 30° in the roll direction, with 5° step.

Finally, two studies were performed to compare the performance of our proposed method with existing techniques (this serves as additional evidence that the proposed method is superior in terms of accuracy and speed).

Results

For each clinical scans the three discrepancy measures α , d and θ_{GT} are calculated. Their values of mean, SD and the maximum discrepancies are given in Table 1.

The distribution of errors (α , d) is given in Figs. 3a and b, respectively. As shown in the plots, the errors are centralized around zero. Among all cases, more than 75% of them have angular error and distance deviation less than 1° and 1 mm, respectively.

We also performed non-parametric (Kendall's τ) as well as parametric (Pearson's linear correlation) tests for checking whether there is any correlation between (i) α and θ_{GT} and (ii) d and θ_{GT} . The results are tabulated in Table 2.

The robustness of the method towards rotation is checked by applying the algorithm to artificially rotated volumes in yaw, pitch, and roll directions. The mean, SD and maximum values of α , d , and θ_{GT} for the rotated volume are tabulated in Table 3. The algorithm is able to extract the MSP up to 40° in yaw, 30° in pitch, and 25° in roll direction accurately.

Fig. 3 Distribution plots of (a) angular discrepancy and (b) distance deviation, notice that errors are centralized around zero

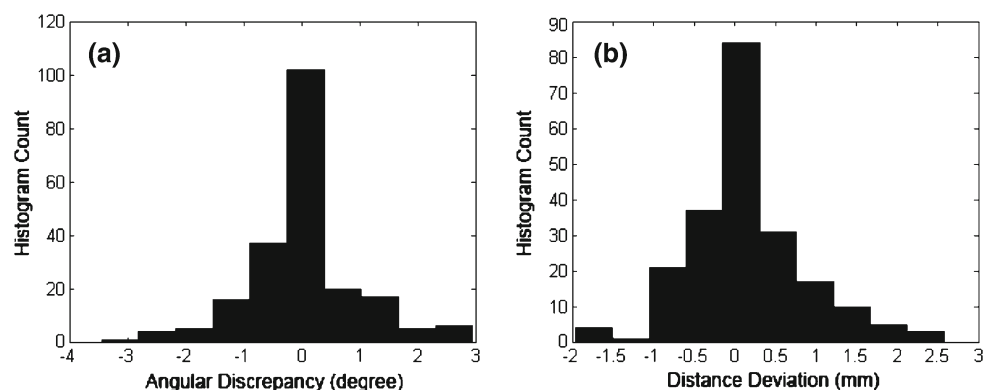


Table 2 Results for testing the existence of a correlation between the variables

	(α, θ_{GT})	(d, θ_{GT})
Kendall's τ	0.3750 (1.345e-11)	0.2917 (1.430e-7)
Pearson's linear correlation	0.4628 (2.193e-9)	0.4739 (8.687e-10)

The quantities in the parenthesis indicate probability of chance fluctuation

Table 3 Statistical parameters of distance and angular discrepancies between the calculated MSP and ground truth MSP for the rotated volume

	α (°)	d (mm)	θ_{GT} (°)
Mean	0.7060	0.6394	0.9094
SD	1.551	1.2787	0.6987
Max	3.4814	2.9262	2.0735

Yaw angle: 5° to 40°, pitch angle: 5° to 30°, roll angle: 5° to 25°

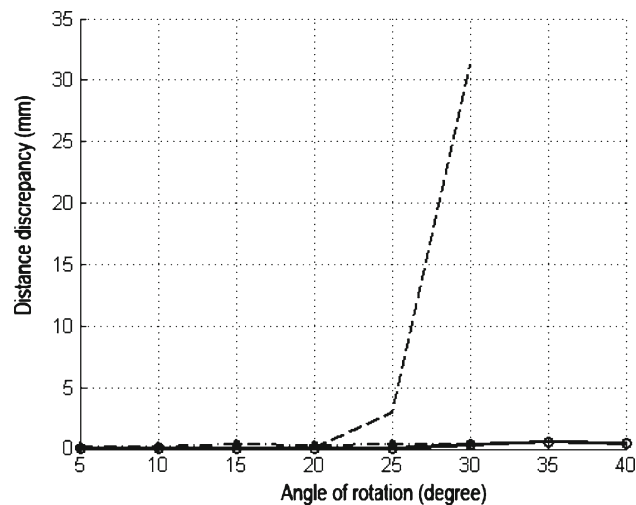
**Fig. 4** The distance discrepancy plotted against angle of rotation (solid, dotted and dashed lines represent distance discrepancy for volume rotated in yaw, pitch and roll directions, respectively)

Figure 4 shows the distance discrepancy plotted against the angle of rotation for the rotated volume. The solid, dotted and dashed lines represent distance discrepancy for volume rotated in yaw, pitch, and roll direction, respectively. From Fig. 4, it is observed that the MSP deviates largely when the roll angle reaches 30°.

Discussion

Our goal is to devise a method for MSP extraction applicable in a clinical setting. Therefore, the method has to be robust, rapid, accurate and automatic and take into account the process of data acquisition. We discuss this method in three

categories: (1) justification of parameters, (2) performance including accuracy, speed, and robustness towards head rotation, and (3) comparison of our method with existing techniques (this serves as additional evidence that the proposed method is superior in terms of accuracy and speed).

Parameter justification

Selection of threshold values for θ_d and lr_ratio

The reason why we use two parameters to decide whether to rotate the original volume is as follows. There are three possible head rotations during CT acquisition: yaw, roll, and pitch. The third rotation only affects the in-plane angle in the MSP and, therefore, our MSP algorithm is not sensitive to this angle. The yaw angle of patient's head is approximated by $\theta_{ellipse}$ and the yaw angular discrepancy between the MSP and the actual IF is captured by parameter θ_d . However, by using θ_d alone, we were unable to capture the roll angular discrepancy of the MSP.

Therefore, we introduce lr_ratio as the second parameter to capture the roll angular discrepancy indirectly. In the axial plane, the roll angle of the patient's head can be visualized by the distance displacement of the IF from one axial slice to the next one, where larger displacement indicates a higher roll angle. If the MSP coincides with the IF, the parameter lr_ratio would be close to one. Conversely, when the MSP is inaccurately identified, so that the roll angle of the MSP is different from the roll angle of the IF, there will be a noticeable distance displacement between MSP and IF in the axial slices. Hence, the lr_ratio would not be close to one.

As mentioned in "Algorithm", it is necessary to rotate the original volume when θ_d is greater than 7.8° or lr_ratio is larger than 10%. These two values were chosen based on eightfold cross validation experiments. Leave-one-out method was applied such that in each experiment seven groups are used as training set and one group as test set.

For each experiment, the best combination of θ_d and lr_ratio threshold value (θ_{d_const} and lr_ratio_{const}) was derived as follows. For each volume in the training set, method [21] was applied to compute the MSP and parameters θ_d and lr_ratio were calculated from the volume. Threshold parameters θ_{d_const} and lr_ratio_{const} were varied within interval of 6–8° and 9–10.5% with the step size of 0.1, respectively. Then, for each combination of threshold parameters θ_{d_const} and lr_ratio_{const} , the Dice's coefficient was calculated as a measure on how good the parameter combination can sort out the successful and failed cases of the MSP computed using method [21].

We selected the best combination of θ_{d_const} and lr_ratio_{const} that gave the best Dice's coefficient for that particular experiment and applied the parameters to the corresponding test datasets. Figure 5 shows the comparison

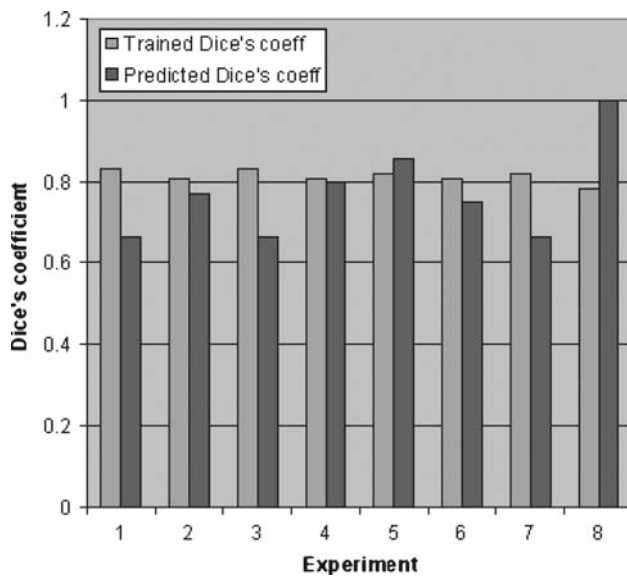


Fig. 5 Comparison of the trained and predicted Dice's coefficient obtained from training and test set for each experiment

between Dice's coefficient obtained from the training and test set for each experiment. The prediction error has mean and SD of -0.0041 and 0.131 , respectively. The parameters derived from the experiments has median of 7.8° (10%) for θ_{d_const} (lr_ratio_{const}) with the SD of $0.13(0.0009)$, respectively.

After performing all eight experiments, we got eight pairs of θ_{d_const} and lr_ratio_{const} (the value of lr_ratio_{const} was the same for all the experiments). From these pairs, we use the median of the obtained θ_{d_const} and lr_ratio_{const} as the final parameters. These final parameters are

$$\theta_{d_const} = 7.8^\circ$$

$$lr_ratio_{const} = 10\%$$

Applying these final parameters to all test sets gives prediction errors with the mean and SD of -0.0012 and 0.131 , respectively.

To check the goodness of the chosen threshold values, we apply method [21] and the proposed method to all datasets and compare the results. Among 208 scans, the method [21] is able to identify MSP accurately in 164 datasets and gives inaccurate plane in the remaining 44 datasets. Using the proposed method, the MSP has been correctly identified for all datasets with the improved accuracy (Table 1).

Selection of KL search range

The performance of the proposed method was analyzed for different values of search range varying from 5 to 40 mm. Figure 6 shows the resulting angular and distance discrepancies for various search range. It is observed that the smallest discrepancy is reached when the search range is 20 mm.

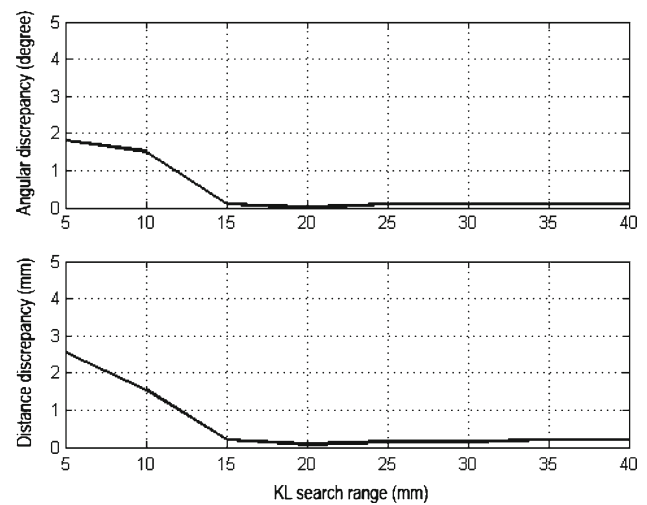


Fig. 6 The plot of discrepancies for KL search range varied from 5 to 40 mm; *top* angular discrepancy ($^\circ$), *bottom* distance discrepancy (mm)

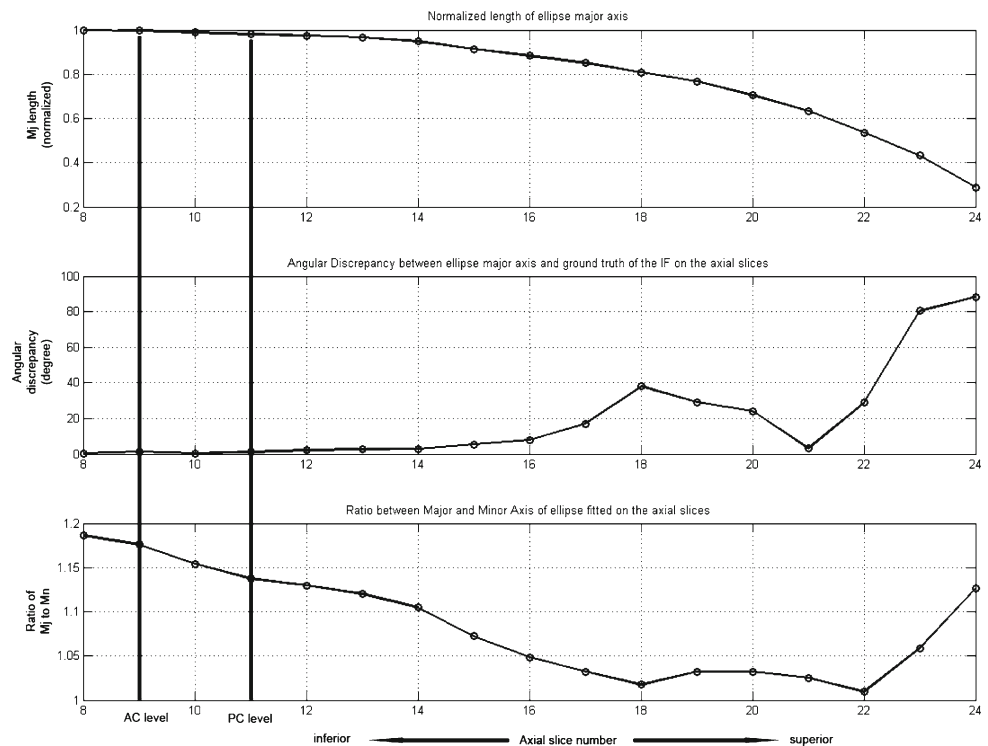
There is no improvement of accuracy when the search range is enlarged beyond 20 mm.

Selection of reference axial slice for ellipse fitting

As mentioned in “Calculation of θ_d ”, the axial plane with the largest anterior–posterior extent is used to fit the ellipse. This step is taken by assuming that in the plane with the largest extent, the skull should have the maximum eccentricity and, therefore, the ellipse M_j should approximate the IF well. The following experiment is conducted to justify this assumption. An ellipse is fitted to the outline of the largest connected component of the skull in each axial slice above the eye level. Three parameters were analyzed: the length of ellipse M_j , the angular discrepancy of ellipse M_j orientation with respect to the IF orientation, and the ratio of semi-major to minor axes length. The result is as shown in Fig. 7. The top, middle, and bottom plots show the normalized length of ellipse M_j , angular discrepancy between ellipse M_j and the IF and the ratio of semi-major to minor axes length, respectively.

In the middle slices (slice 8–13 in Fig. 7), the length of M_j does not vary much (top plot). In this range, the orientation of M_j gives a good approximation of the IF, which is indicated with low angular discrepancy (middle plot). When the length of M_j is maximum (slice no 8), the ratio of semi-major to minor axes lengths (bottom plot) reaches its maximum and the minimum value of angular discrepancy is obtained. This observation is consistent with our previous assumption that the major axis should approximately correspond to the direction of the IF when the skull has the maximum eccentricity in the axial level with largest anterior–posterior extent.

Fig. 7 Top plot Normalized length of ellipse major axis, middle plot angular discrepancy between IF and ellipse M_j , bottom plot ratio of the semi-major to minor axis length of ellipse; all are plotted against the axial slice number



From Fig. 7, it is also observed that as the axial slice approaches the superior level of the brain, the length of M_j decreases, suggesting that the skull size is becoming smaller. Additionally, the angular discrepancy between the actual IF and M_j starts increasing and vary unpredictably. This trend can be explained as follows. As the axial slice approaches the superior level of the head, the skull becomes smaller and more circular, and at the superior-most level, the skull is pear-shaped. Consequently, the ellipse changes in terms of size (length of M_j) and its eccentricity and becomes more circular. Hence, the ratio of the semi-major to minor axes length is close to 1 and the direction of the ellipse axes can be in any arbitrary vector. Additionally, the presence of dura matter and the transition from the brain tissue to the skull in the superior level of low-resolution scans introduces a large partial volume effect which can affect the accuracy of the skull extraction. Therefore, in the superior level, the orientation of ellipse M_j may not approximate the IF accurately.

It is also observed from Fig. 7 that the ratio of semi-major to minor axes lengths in the top-most slices increasing from 1.02 to 1.12 (slice 22–24), suggesting that the skull shape changes from circular to elliptical again. However, the angular discrepancy remains increasing. This trend occurs due to the following reason. When the skull shape changes from circular to elliptical, what actually happens is that the length of M_j decreases until its length is almost equal to the length of ellipse minor axis (M_n); then it continue to decrease until M_j is shorter than M_n , such that the orientation of ellipse

M_j is now almost perpendicular to the IF; thus giving a large angular discrepancy.

Accuracy, robustness and speed

In this study we are dealing mostly with low-resolution CT scans. It is characterized by a large slice thickness whereby it leads to a prominent partial volume effect. This partial volume effect introduces blurring and difficulty in delineating the exact boundaries between various brain structures/tissues. Thus, it poses challenge as to how to detect the IF (for the MSP extraction) in such scans with a large partial volume effect.

Our validation on clinical CT scans shows the following results. The extracted MSP has mean angular errors of $0.0258^\circ \pm 0.9541^\circ$ and distance errors of 0.2944 ± 1.4747 (in pixels) or 0.1472 ± 0.7373 (in mm). This value is better than our previous result validated against MR scans which gave us the average distance and angular deviation of 1.25 pixels and 0.63° , respectively [21], with an additional increase in robustness towards head rotation and high slice thickness.

Figures 8 and 9 show the MSPs projected on various clinical datasets. Figure 8 illustrates the results of MSP calculation for a highly rotated brain. Figure 8a shows the MSP calculated using the method [21] projected on the axial plane. It is noticeable that the calculated MSP does not pass through the IF and has a high discrepancy in its orientation. By incorporating the correction of head orientation in the proposed

Fig. 8 The calculated MSP projected to the axial plane: **a** using [21], **b** using the proposed method, **c** ground truth marked by an expert

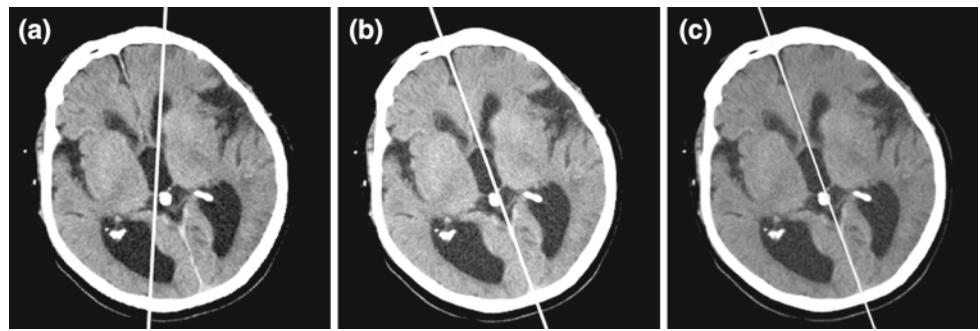
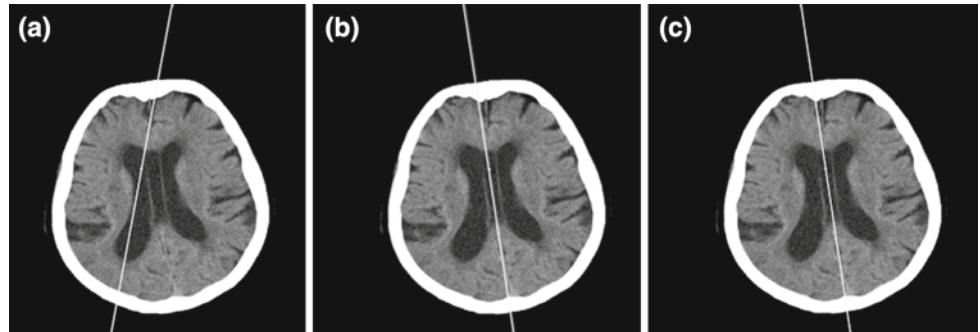


Fig. 9 The calculated MSP on brain with the presence of the intermediate ventricle: **a** using [21], **b** using the proposed method, **c** ground truth marked by an expert



method, the new MSP is as shown in Fig. 8b. Notice that the MSP is not exactly symmetrical with respect to the third ventricle because of the ventricular lateral shift (i.e., the MSP follows the falx cerebri).

Figure 9 shows the result of MSP calculation for another tilted brain volume where the intermediate ventricle is present. Figures 9a and b show the MSP projection on the axial plane calculated using [21] and the new method, respectively. Although the IF is not clearly visible in this case (as compared to the volume shown in Fig. 8), the proposed algorithm is still able to extract the MSP accurately.

The proposed algorithm is applicable to low-resolution scans as well. The method is able to provide an accurate result for low contrast scan even where the IF is not clearly visible and gray and white matter is poorly distinguishable (Fig. 10).

Figures 11 and 12 show the box-and-whisker diagrams of (i) angular deviations from planarity θ_{GT} versus angular discrepancy α and (ii) angular deviations from planarity θ_{GT} versus distance discrepancy d , respectively. The datasets were sorted into six groups based on the value of θ_{GT} . Each group is represented as a box with three lines (lower bound, midline and upper bound) indicating the first quartile, median, and the third quartile values, respectively. The lower and upper whiskers point to 1.5 inter-quartile ranges (IQR) from the first and third quartiles, respectively. Any data which lie more than 1.5 IQR lower than the first quartile or 1.5 IQR higher than the third quartile is considered an outlier (shown as '+' sign). It is clear from Figs. 11 and 12 that as

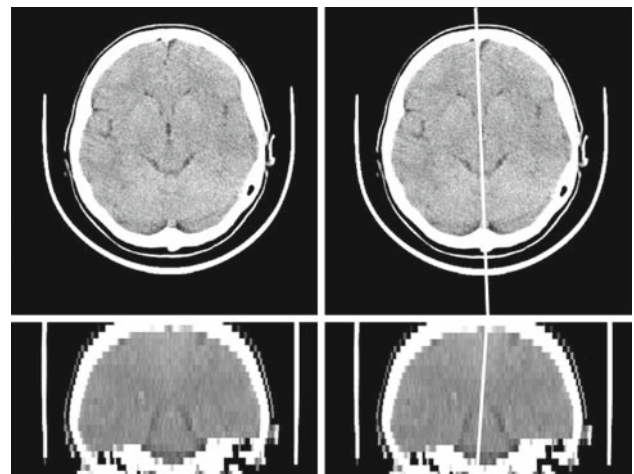


Fig. 10 The application of our method to low contrast CT scan; *left column* original axial and coronal slice, *right column* axial and coronal slice with the resulted MSP projected on it

the θ_{GT} gets higher, the angular discrepancy α and distance discrepancy d have a tendency to increase.

This is consistent with correlation test that we performed which also shows that there is a significant correlation between α and θ_{GT} as well as between d and θ_{GT} as the probability of chance fluctuation is less than 0.05 for both cases (Table 2). Therefore, in datasets where the value of θ_{GT} is high, which implies that the ground truth lines are not parallel to each other, we can expect that the angular and distance discrepancies tend to have higher values as well.

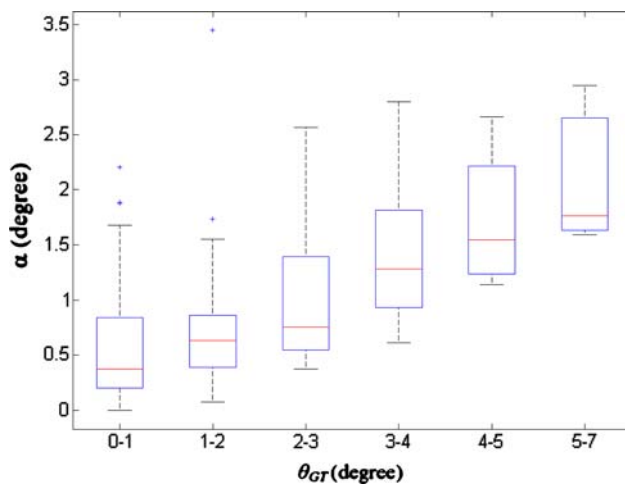


Fig. 11 The box-and-whisker diagram of angular deviations from planarity θ_{GT} versus angular discrepancy α . Outliers are denoted by *plus* symbol. Note that as the θ_{GT} gets higher, the angular discrepancy α has a tendency to increase

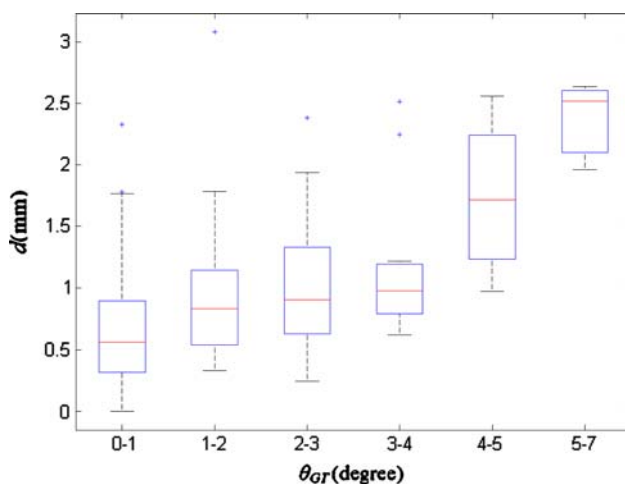


Fig. 12 The box-and-whisker diagram of angular deviations from planarity θ_{GT} versus distance discrepancy d . Outliers are denoted by *plus* symbol. Note that as the θ_{GT} gets higher, the distance discrepancy d has a tendency to increase

When we investigate the cases with high discrepancies (especially those denoted as outliers in Figs. 11 and 12), we found that the IF is curved (not planar), as illustrated in Fig. 13. In this case, the calculated MSP accurately approximates the IF in the posterior region of the brain, but giving a significant discrepancy in the anterior part. Conversely, the ground truth line approximates the IF accurately in the anterior part, with compensation of slight deviation in the posterior region.

In normal brains with heavily curved IF, the MSP may not give the best approximation of the IF since the IF itself is a curved surface. Similarly, the proposed MSP algorithm may not give the best approximation of the IF in pathological cases with the presence of midline shift (where the

midline of the brain is displaced due to massive swelling of one hemisphere). Part of our future study would be to explore a solution how to best approximate the IF in such situations.

The robustness of the algorithm towards rotation was validated using artificially rotated volume in yaw (5° to 40°), pitch (5° to 30°) and roll (5° to 30°) direction with 5° step. We do not test higher rotation angle for pitch direction since our algorithm is not sensitive to changes in this angle. In the case of roll angle, we stop the testing at 30° where the method gives large discrepancy (more than 3 cm), see Fig. 4.

From the result of method applied to artificially rotated volume, it is observed that the mean angular and distance deviations are less than 1° and 1 mm for all cases (Table 3), except for the volume rotated 30° in roll direction (which is very rare, see below), that gives large distance discrepancy. Hence, our proposed algorithm is able to extract the MSP accurately for scans with roll angle less than 30° .

To check the applicability of our method in the clinical setting, we analyze the degree of head tilt in the clinical datasets. For each clinical scan, the MSP ground truth was generated using the least square error method from the MSL marked by the neuroanatomy experts. The yaw and roll angle of the MSP ground truth with respect to the vertical plane was calculated (note that we did not calculate the pitch angle as its calculation requires additional ground truth of AC and PC landmarks). It was observed that the yaw and roll angles of the scans vary from 0° to 35° and from 0° to 17° , respectively. For the yaw angle, about 85% of the datasets have the yaw angle less than 10° , 10% between 10° and 15° , and 5% between 15° and 35° . As for the roll angle, about 94% of them have the roll angle less than 10° , less than 6% of them between 10° and 17° , and none of them has roll angle larger than 17° . And for these clinical cases where the roll angle is larger than 10° , the algorithm is still able to extract the MSP with mean angular errors less than 2° and distance errors less than 1 mm.

Rotation of the brain in roll direction affects visual brain symmetry in the CT axial slice. When the roll angle is large, it is hard to analyze the brain structures since they may have different manifestations in the left and right hemispheres. The proposed method may work incorrectly when the roll angle is above 30° . However, given that in the clinical setting it is rare to have roll angle more than 10° (less than 6% of the analyzed clinical datasets have roll angle between 10° and 17° , and no dataset has it above 17°), it implies that it is unlikely to have the roll angle equal or larger than 20° . Thus, the method proposed is still applicable for use in the clinical setting.

This method has been implemented in Microsoft Visual C++ and works in a fully automatic way. For example, in a 24-slice patient's dataset, the algorithm takes about 10 s in 3 GHz CPU with 1 GB RAM.

Fig. 13 MSP projection with high discrepancies (a) axial slice with a curved IF (b) ground truth line (c) projection of the calculated MSP. High discrepancy occurs when the IF itself is a curved surface, hence the MSP may not give the best approximation of the IF

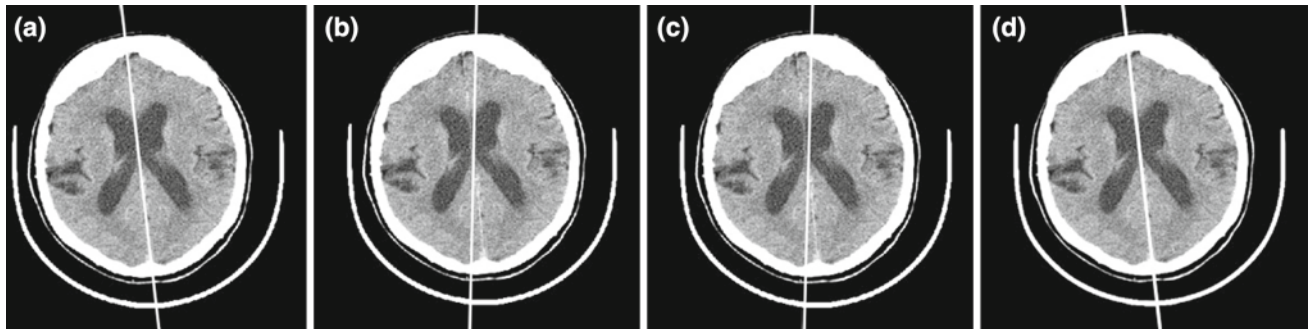
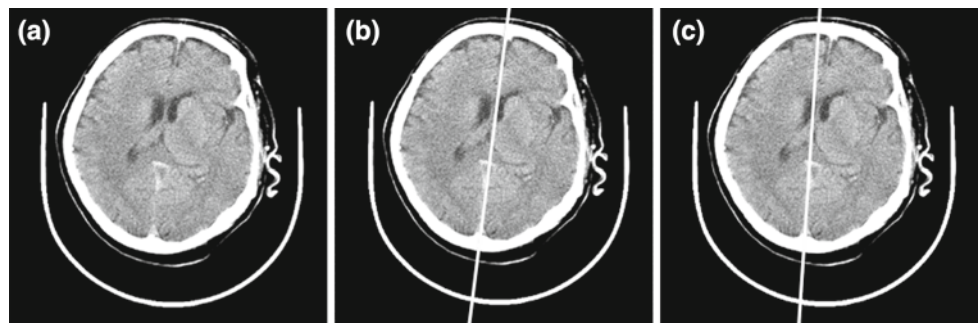


Fig. 14 MSP projection on the axial plane: **a** ground truth; **b** intensity-based cross-correlation; **c** edge-based cross-correlation; and **d** using the proposed method

Comparison with other techniques

Comparison with cross-correlation technique for MSP extraction

We compared the performance of our proposed method with the existing method that is based on cross-correlation technique (see Fig. 14).

The cross-correlation was calculated in the similar manner as described in [7]. Given the patient's image, the midsagittal line was marked in each axial slice by an expert. The axial slice was then cut into two parts along to the midsagittal line to obtain the two hemispheres and cross-correlation was computed between the two images. Afterwards, the middle point in the IF was marked and the line passing through this point was rotated about the marked point in varying angle ranging from -40° to 40° from the original position (of the MSP) and cross-correlation was computed. The lines that give the maximum value of cross-correlation were shown in Figs. 14b and c. They do not accurately coincide with ground truth marking of the IF, as compared with the result of our proposed method (Fig. 14d). Additionally, the cross-correlation based method takes longer time to get the results (approximately 7 min for 24-slice brain image), which is undesirable for usage in clinical setting, especially under emergency conditions. This shows that our proposed method is more superior in terms of accuracy and speed.

Comparison with principal axis method

In the initial stage of our algorithm, the yaw angle of the scan is estimated with the aid of elliptical fitting in the reference axial slice. We compared the performance of our elliptical-based method with principal axis method to estimate the yaw angle of the scan. First the binarized volume is obtained to exclude non-brain structures. This volume is used to calculate the covariance matrix, from which the three axes are obtained. Figure 15 shows the result of our experiment: the major axis (M_j) of the ellipse approximates the IF much better than the principal axis method.

This experiment result is consistent with [15] where an optimization technique to minimize the inaccuracy of the principal axis method is proposed.

Conclusion

A robust MSP extraction algorithm was developed and validated in this study. It extends our earlier method [21] to be applied to CT scans with high slice thickness, large partial volume effect, with the presence of substantial head rotation. Under the proposed method, the correction of head orientation is incorporated by estimating the coarse head orientation using skull-based elliptical fitting on the axial plane and reorienting the volume for better initialization.



Fig. 15 Comparison between principal axes method and ellipse-based method

The algorithm has been validated on 208 low-resolution clinical CT cases (including normal and pathological sets) with rotation angles up to 35° and 17° in yaw and roll direction, respectively. The calculated MSP has been validated against the ground truth marked by neuroanatomy experts and the quantitative analysis shows that the mean of angular and distance errors are less than 1° and 1 mm, respectively. It is also validated on artificially rotated high-resolution scan with the rotation angle up to 40° , 30° , and 25° in yaw, pitch and roll directions, respectively, and the mean of angular and distance errors less than 1° and 1 mm, respectively.

The algorithm is fully automatic, fast (completed within 10 s in average), robust, gives high accuracy and therefore is suitable for clinical applications, particularly in ER. This algorithm has been implemented as a part of our stroke CAD system for ER [28], which was presented at RSNA 2007, ischemic stroke CAD, and is in the process of incorporation into a hemorrhagic stroke CAD [29] (to be used in a clinical trial, phase 3).

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